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June 6, 2026 – Paestum, Salerno

Introduction to OpenFOAM® Computational Library and Design of Profile Extrusion Dies

P3 - Hands-On Profile Extrusion Dies

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JKU IPPD Institute of Polymer Processing and Digital Transformation

JKU JOHANNES KEPLER UNIVERSITY LINZ

Outline

10:00 – 11:15	**Introduction to OpenFOAM and Simulation Fundamentals (P1)**
11:30 – 13:00	**Geometry, Mesh Generation, and Case Setup in OpenFOAM (P2)**
13:00 – 15:00	Lunch break
15:00 – 17:00	**Hands-On Profile Extrusion Dies (P3)**

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Session Contents

- Presentation
- Geometry generation (FreeCAD)
- Mesh generation
- Solver *hpc4PEFoam*
- Bird Carreau + Arrhenius Constitutive model
- Objective Function and convergence
- Run and test

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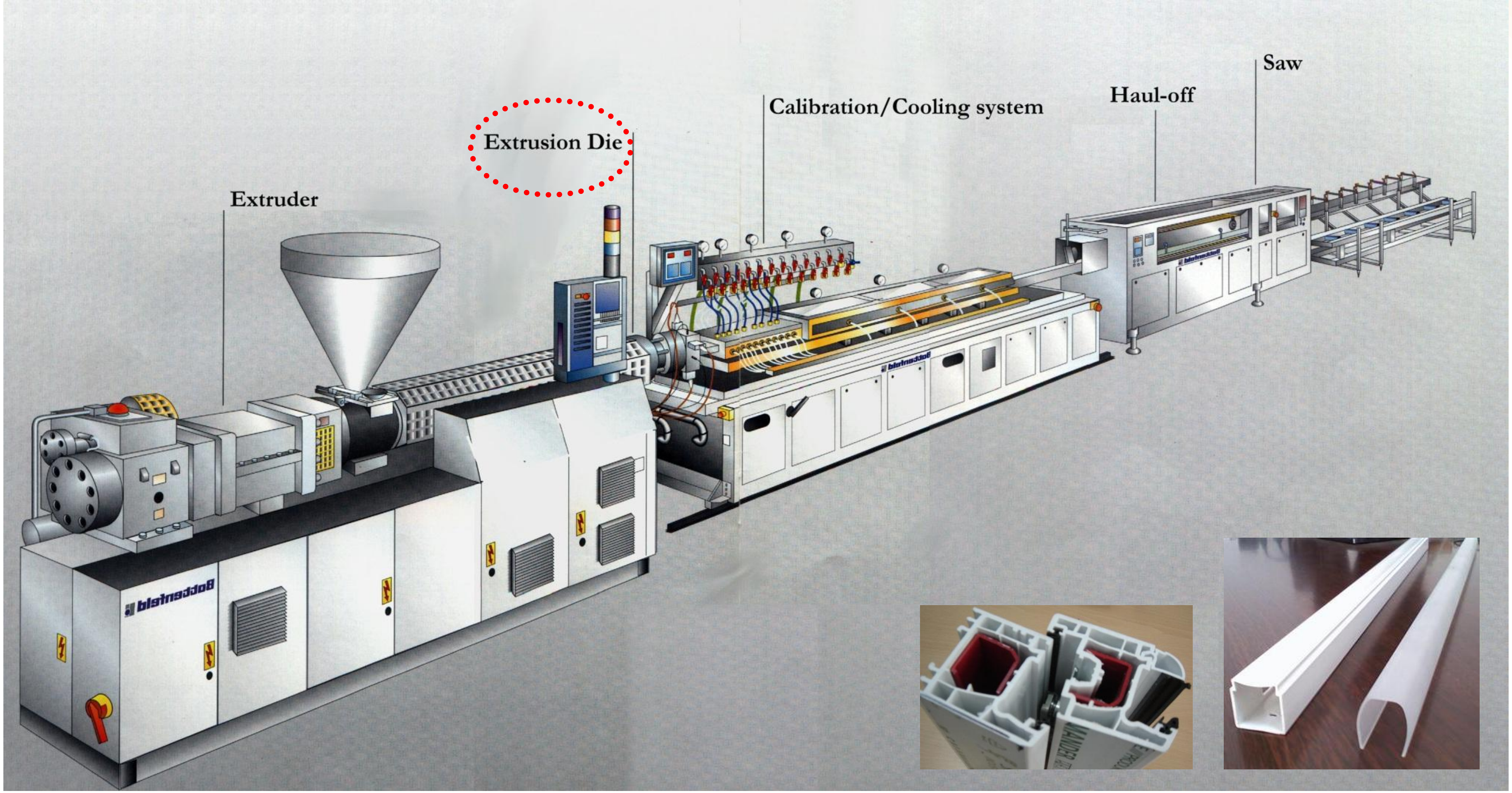
Introduction



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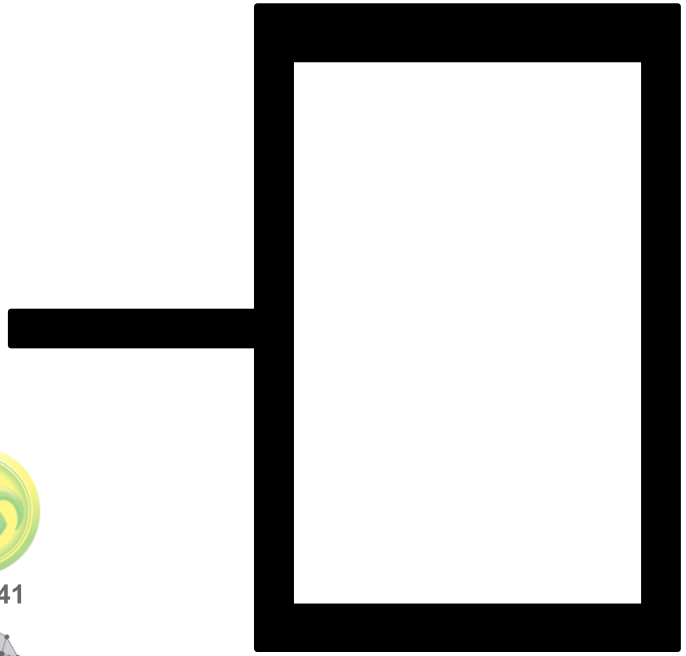
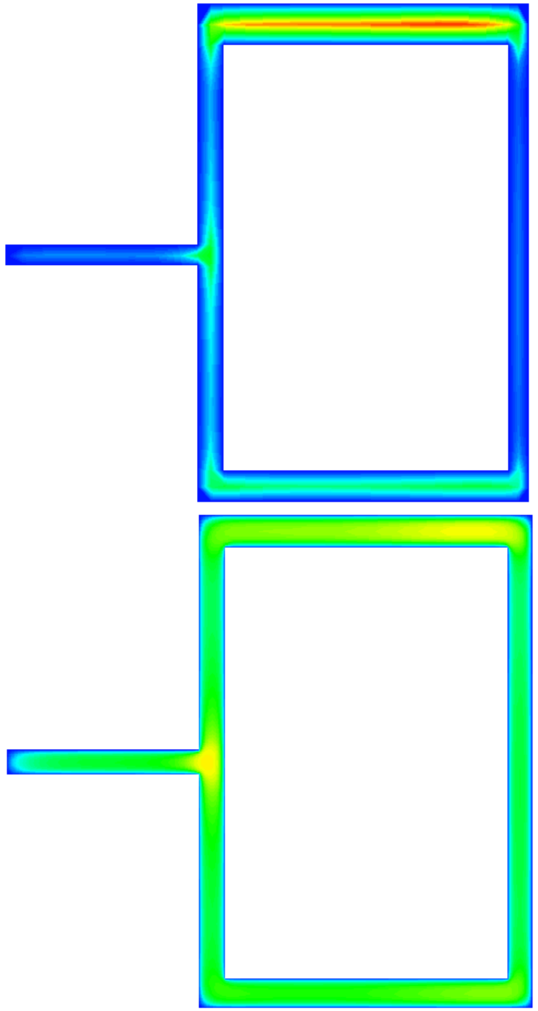
Introduction – Flow Balance

Die land cross-section



Velocity contours



Extrusion run

Unbalanced

Balanced

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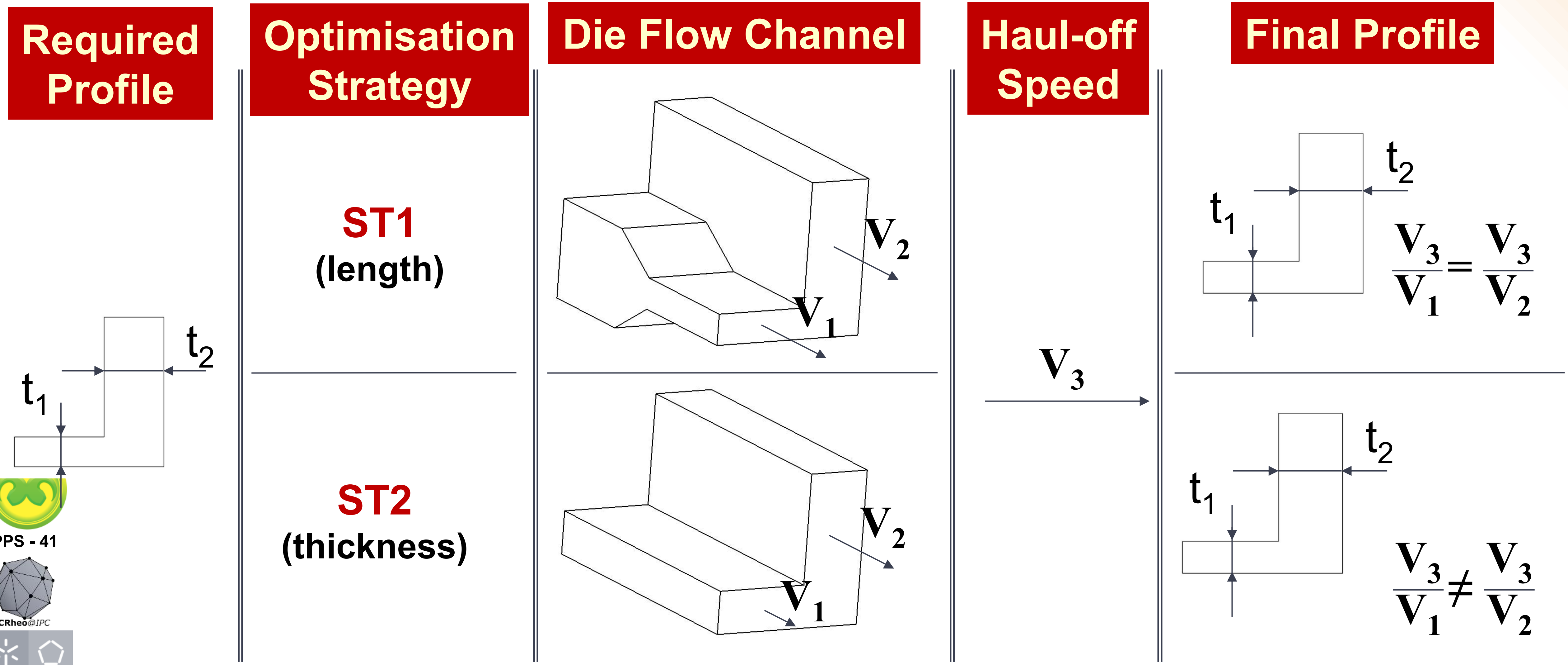
OS Carneiro, JM Nóbrega (Editors) , Design of Extrusion Forming Tools, Smithers Rapra Technology, ISBN 9781847355171, 2012.
JM Nóbrega, OS Carneiro, FT Pinho, PJ Oliveira International Polymer Processing, 2004

Introduction – Optimisation Strategies



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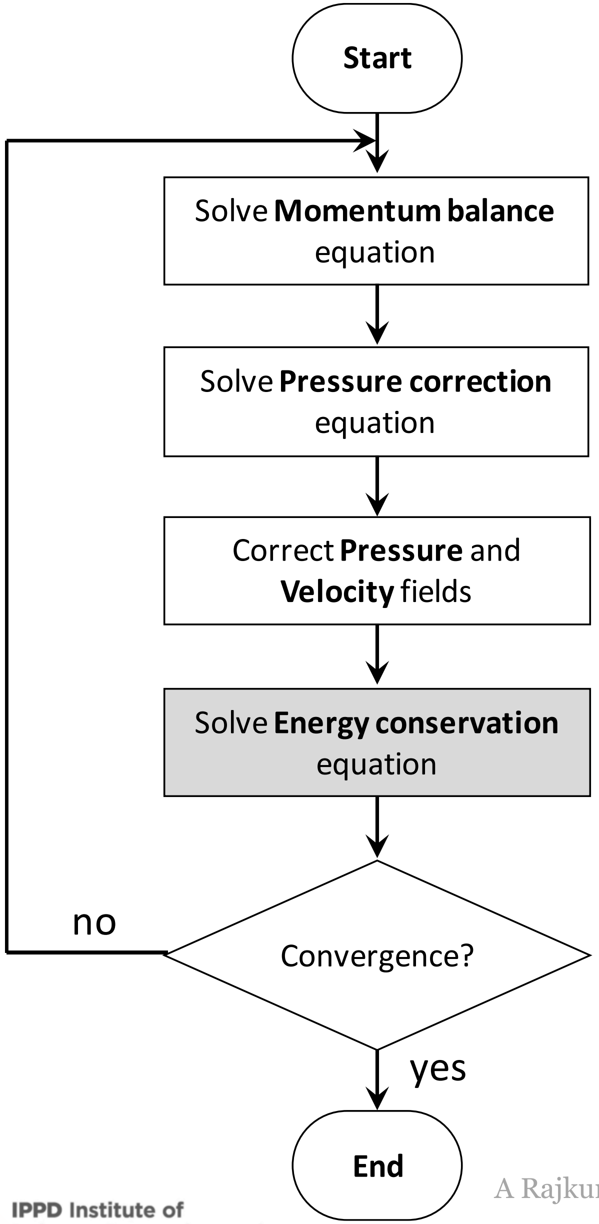
Implementation in OpenFOAM



Methodology Employed

Energy conservation equation (Steady State)

$$\nabla \cdot \left(\boldsymbol{U} T \right) - \nabla \cdot \left(\frac{k}{\rho c_p} \nabla T \right) = \frac{1}{\rho c_p} \boldsymbol{\tau} : \nabla \boldsymbol{U}$$

OpenFOAM Code

```
Solve
 (
      fvm : : div (phi , T)
    - fvm : : laplacian (DT , T)
  == (1 / cp) * (tau  &&  grad U)
)
```

A Rajkumar et al., An Open-source framework for the CAD of Complex Profile Extrusion Dies, International Polymer Processing, 33, 2018

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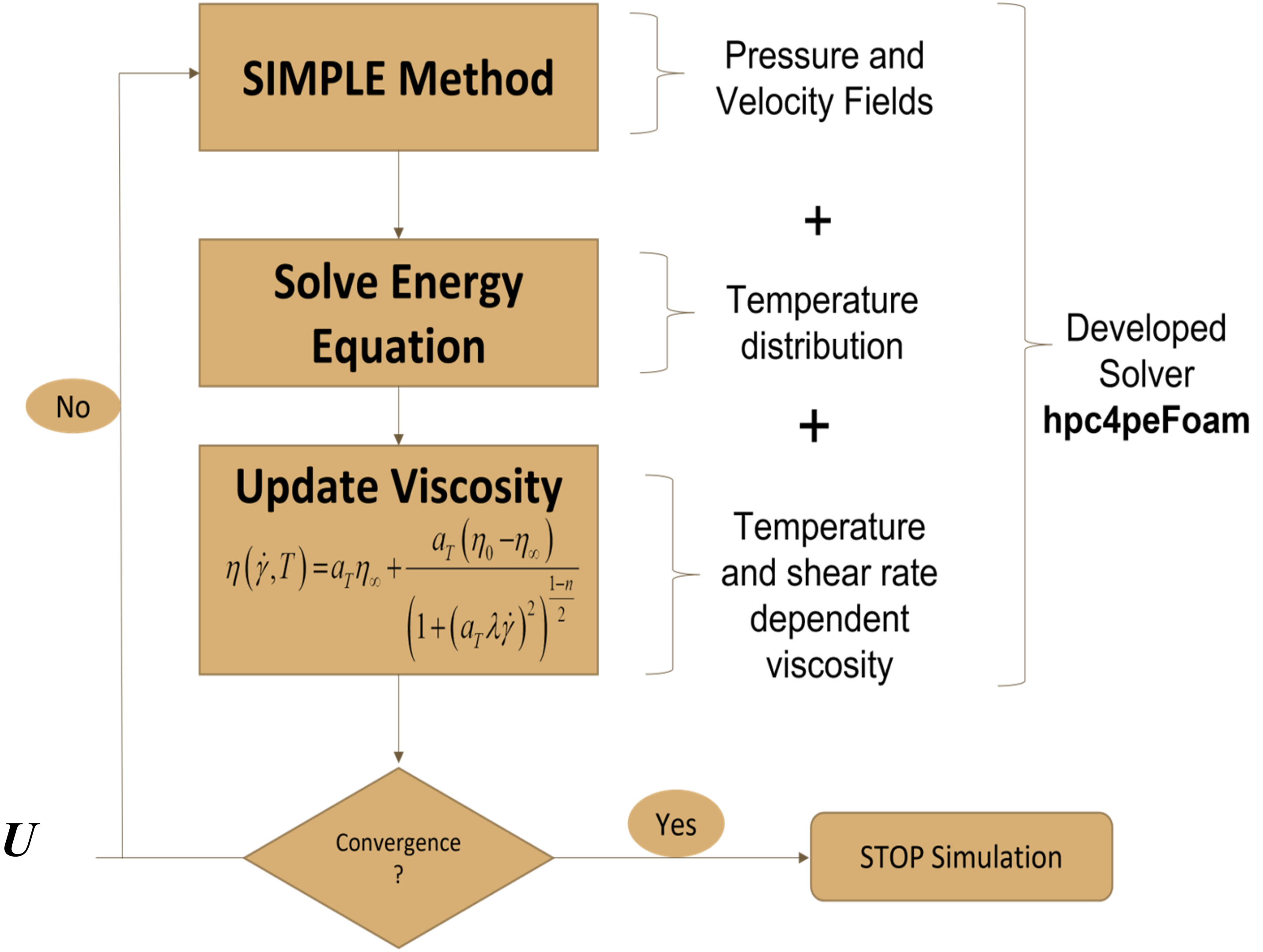
hpc4PEFoam

Simulation Code

OpenFOAM

Energy conservation equation (Steady State)

$$\nabla \cdot (UT) - \nabla \cdot \left(\frac{k}{\rho c_p} \nabla T \right) = \frac{1}{\rho c_p} \boldsymbol{\tau} : \nabla \boldsymbol{U}$$



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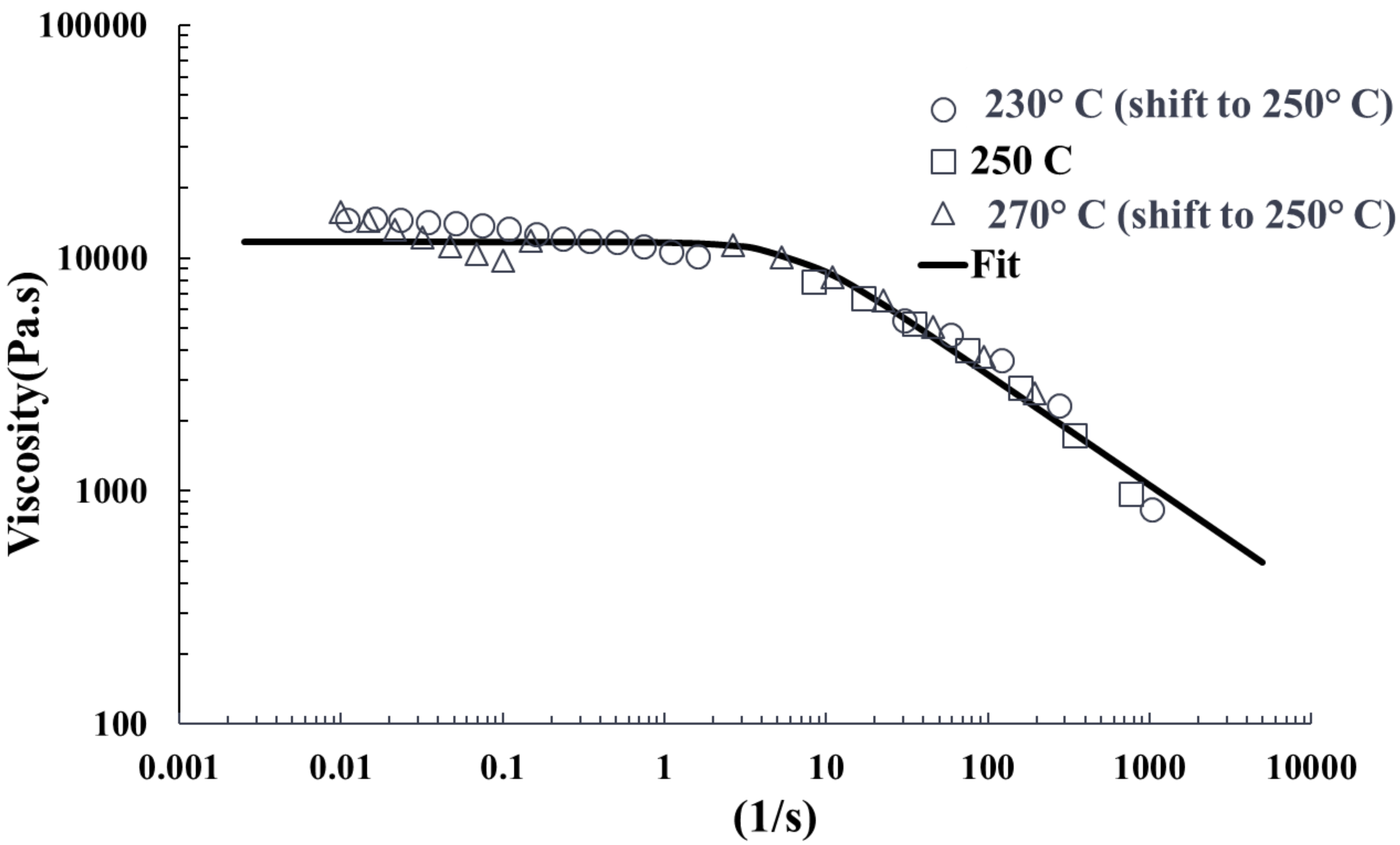
Implementation in OpenFOAM

Material characterization



Polycarbonate

Bird Carreau + Arrhenius Law

$$\eta(\dot{\gamma},T) = a_T\eta_\infty + \frac{a_T(\eta_0 - \eta_\infty)}{\left(1+(a_T\lambda\dot{\gamma})^2\right)^{\frac{1-n}{2}}}$$

Shift Factor

$$a_T = \exp\left(\frac{E}{R}\left(\frac{1}{T} - \frac{1}{T_0}\right)\right)$$

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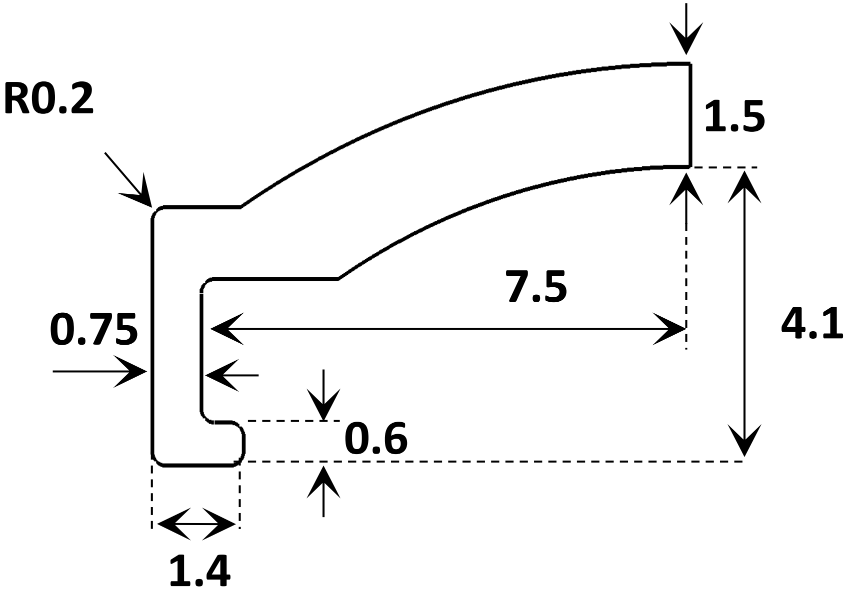
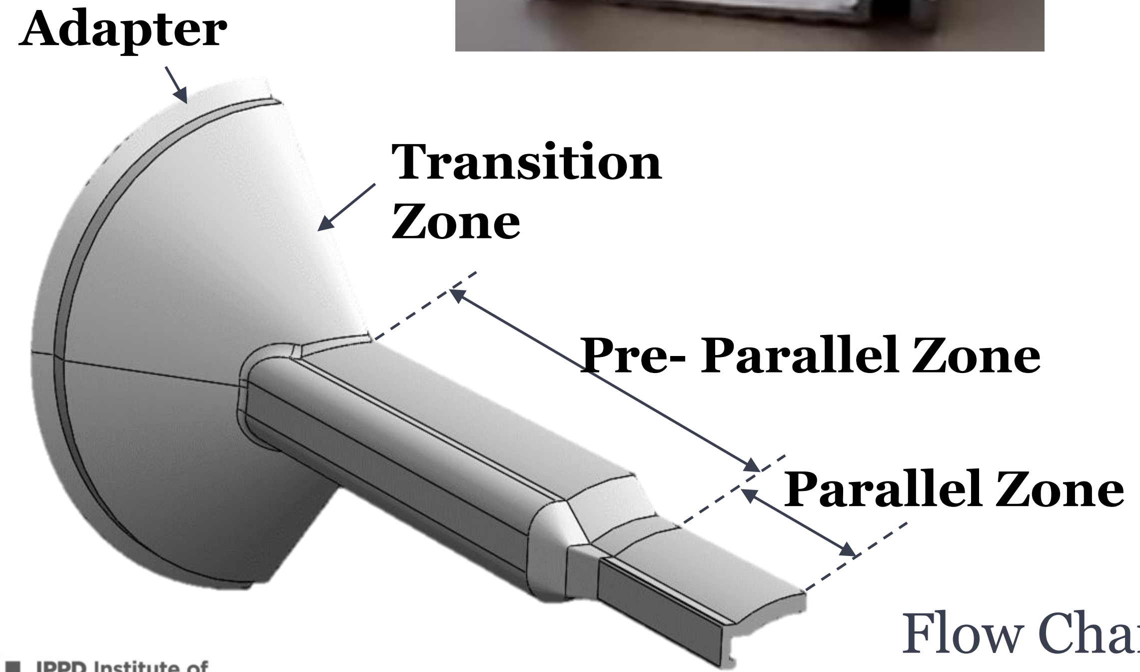
Implementation – Case Study



Profile Cross Section (*dimensions in mm*)



Flow Channel Geometry

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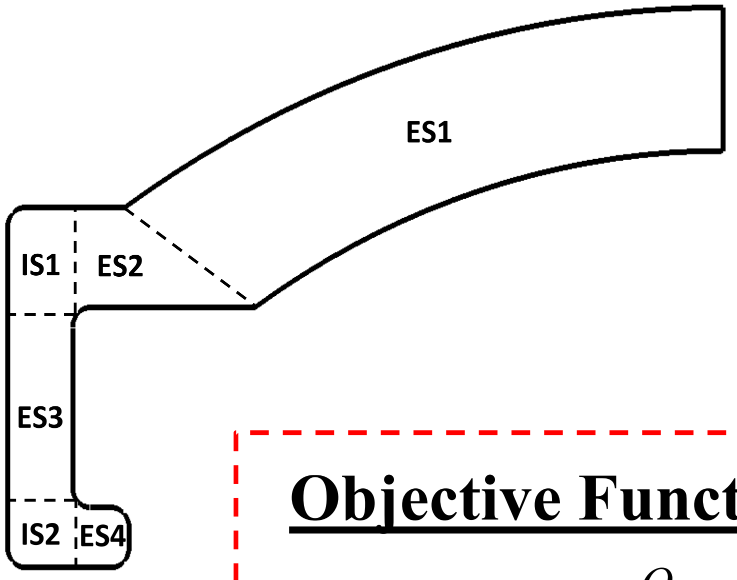
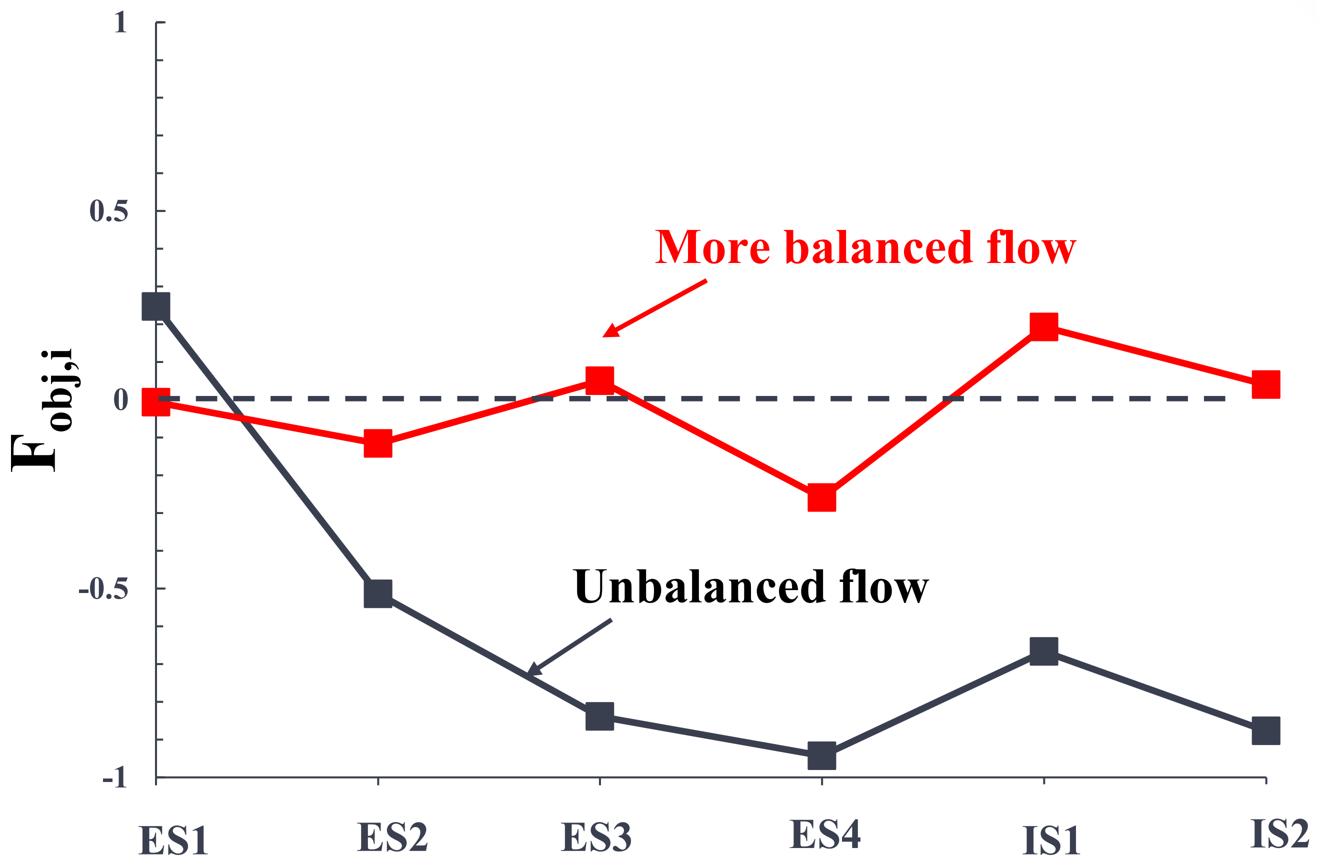
Implementation – Case Study



Objective Function

$$F_{obj,i} = \frac{\frac{Q_i}{Q_{\text{target,i}}} - 1}{\max\left(\frac{Q_i}{Q_{\text{target,i}}}, 1\right)}$$

$$F_{obj} = \frac{\sum_{ES+IS} \left\| F_{obj,i} \right\| A_{\text{target},i}}{A_{\text{target},tot}}$$



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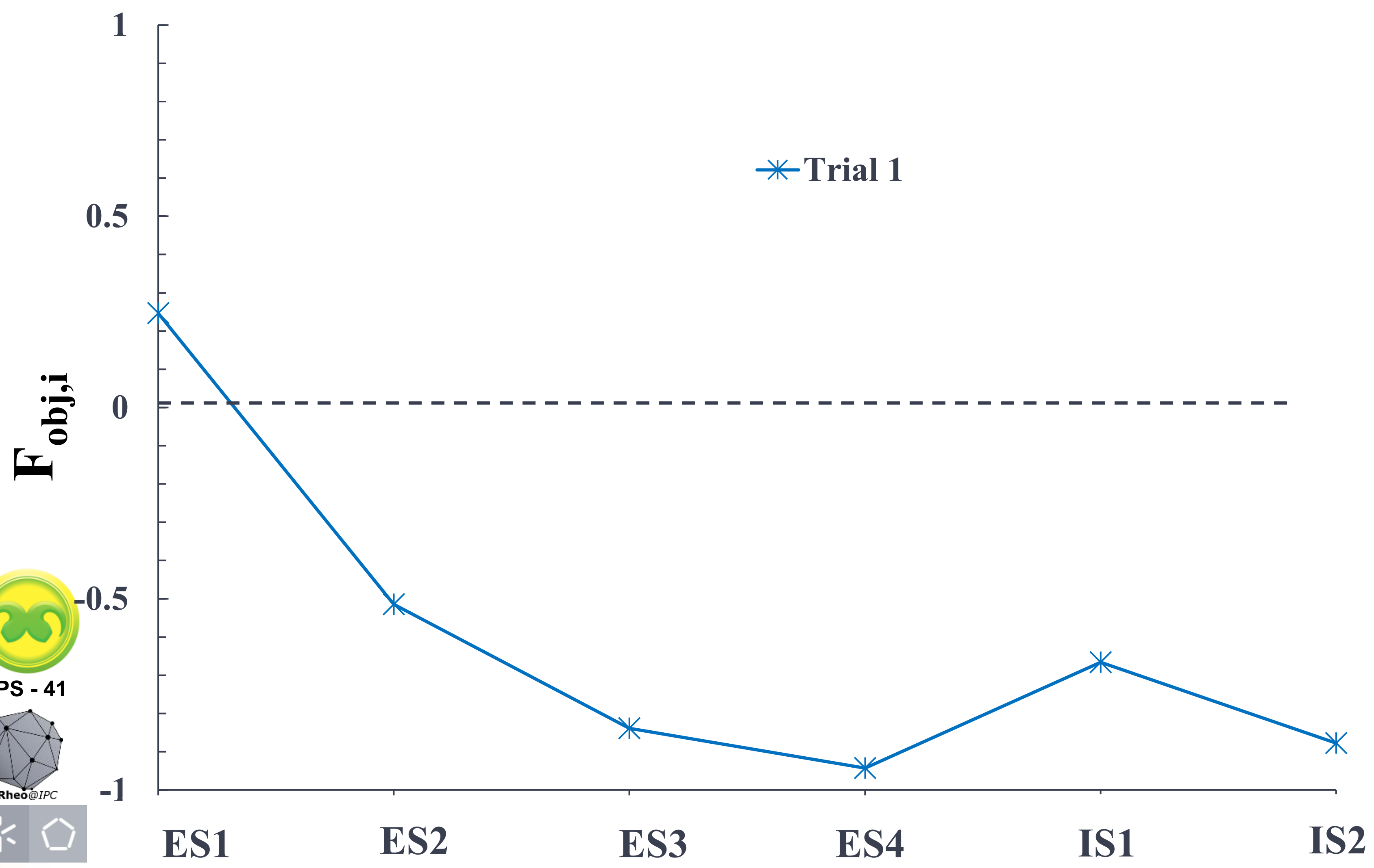
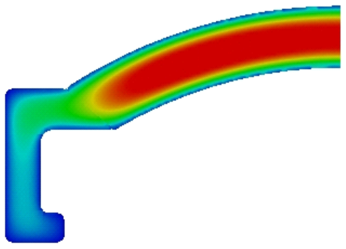
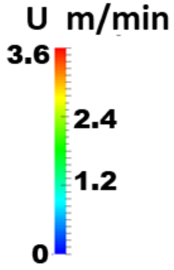




Trial 1



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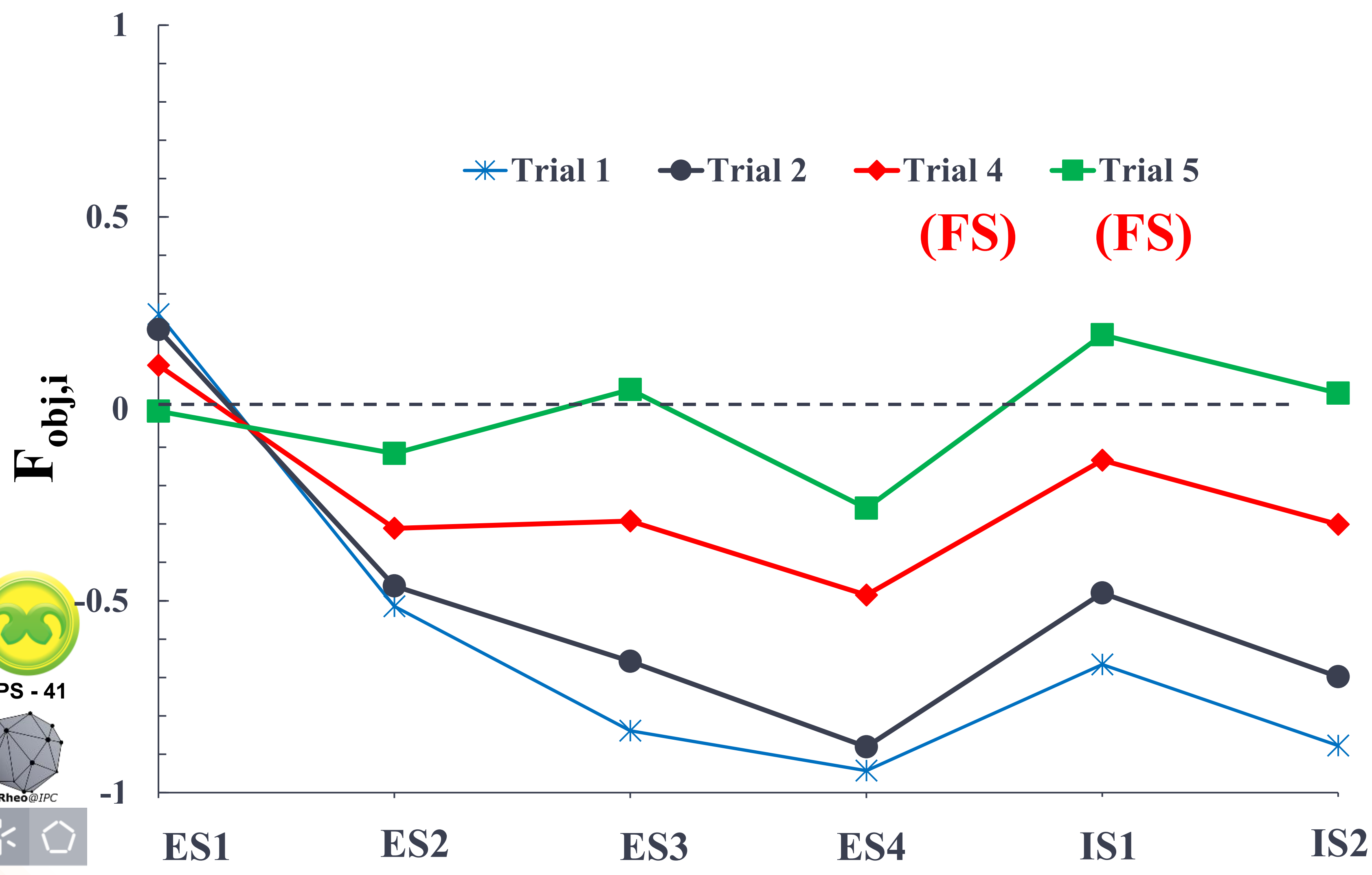
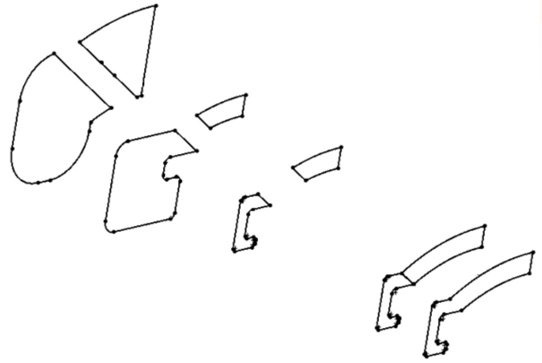
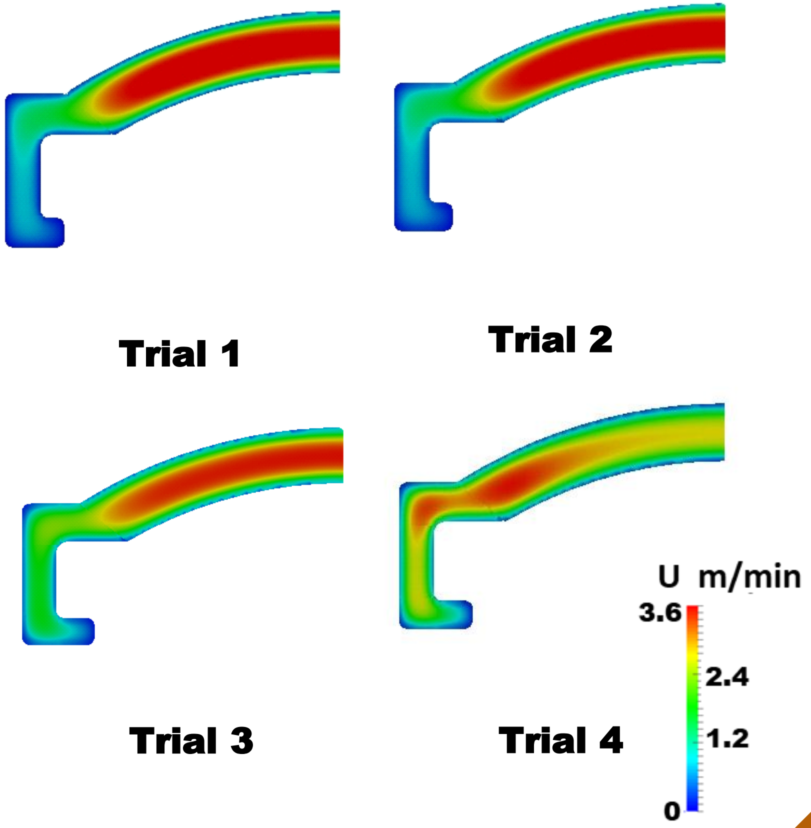


Flow Separator (FS)





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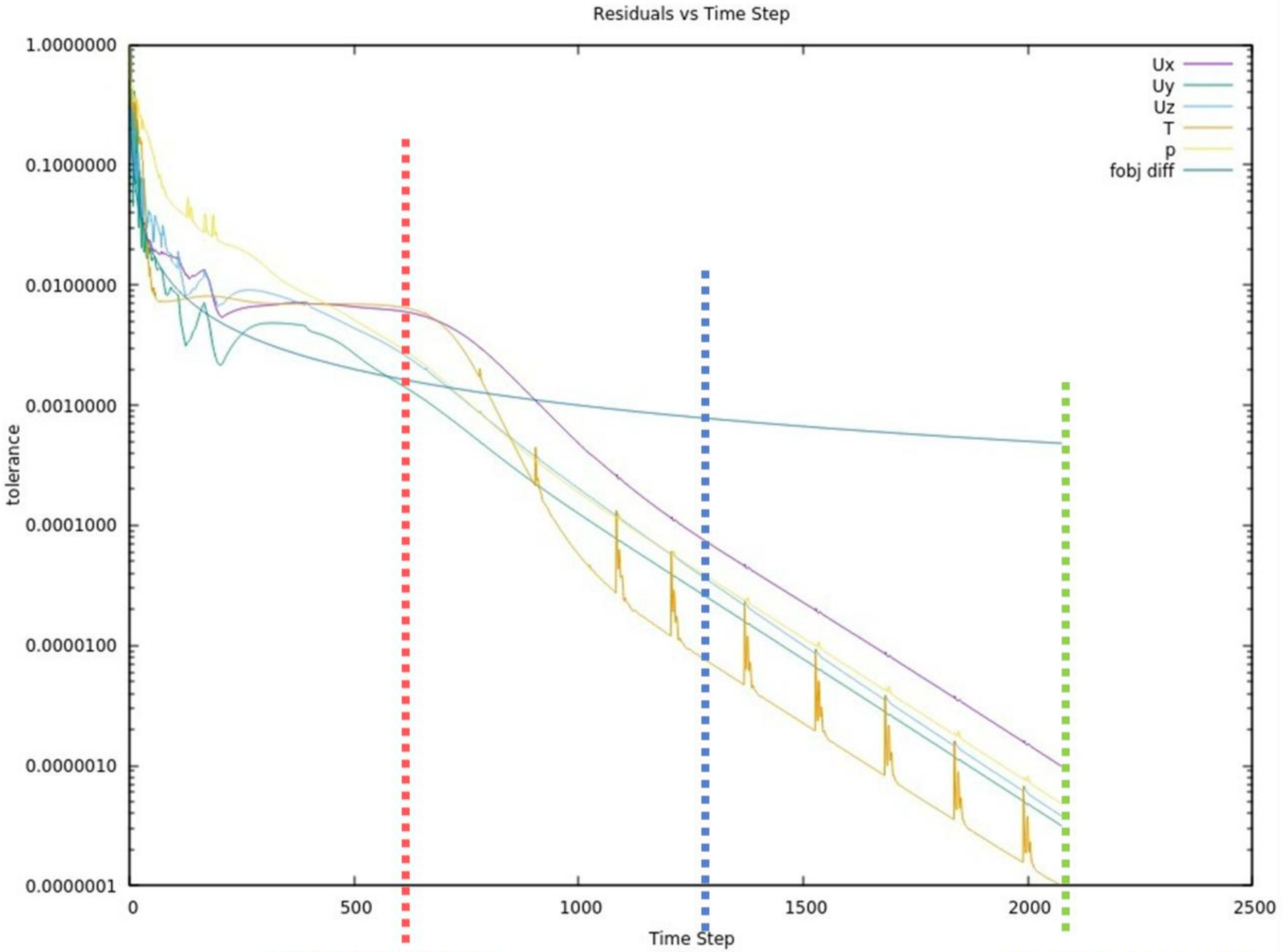
Convergence Criteria

Based of the F_{obj} variation



Simulation stops when Objective function stabilizes (tol < 1%)

Simulation stops based on Residuals (U, p and T converge when reach 10^{-4})

Simulation stops based on Residuals (U, p and T converge when reach 10^{-8})

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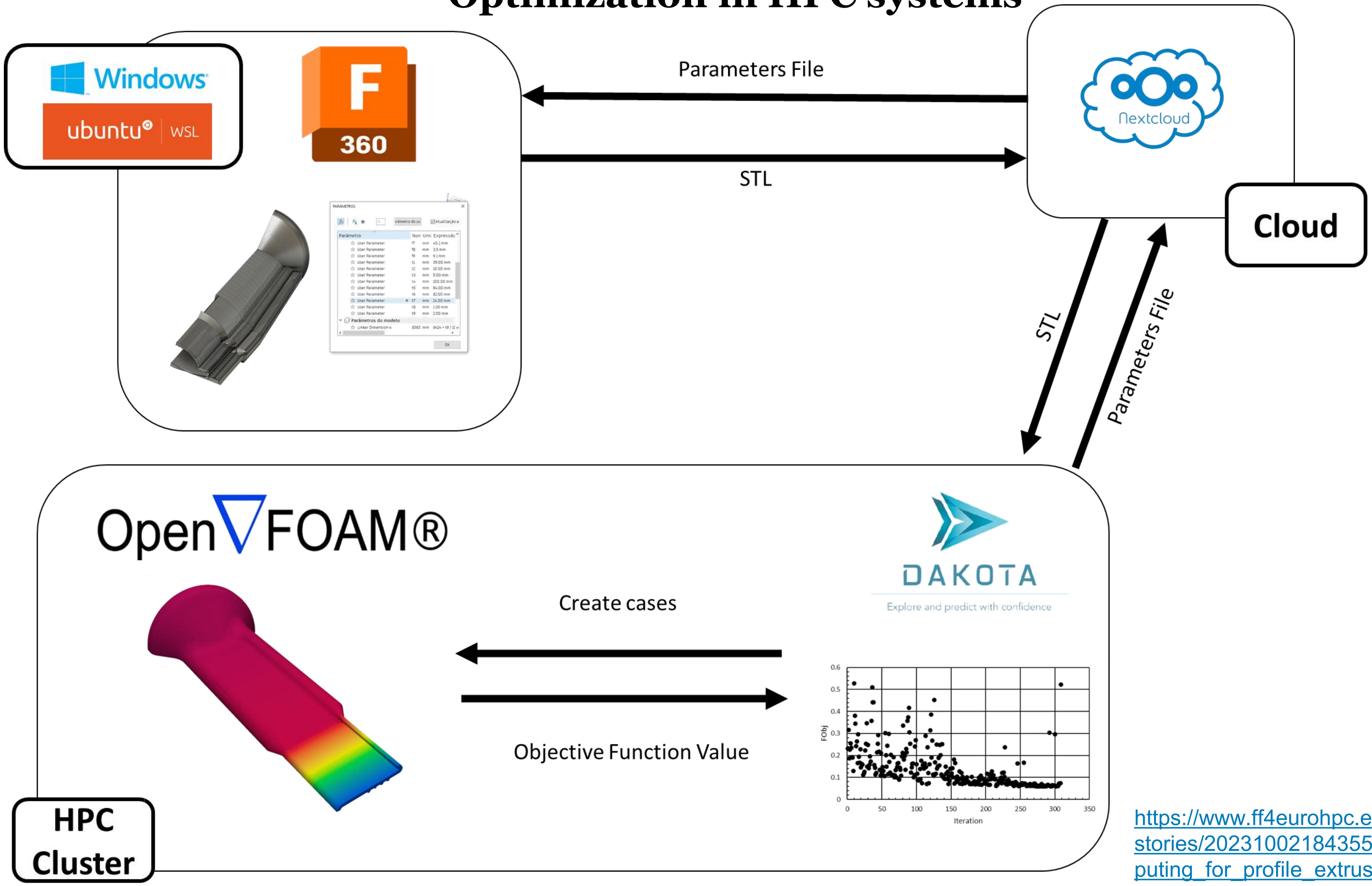
hpc4PE – Calculation Framework

Optimization in HPC systems



https://www.ff4eurohpc.eu/en/success-stories/2023100218435592/high_performance_computing_for_profile_extrusion

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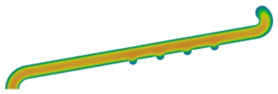
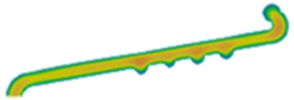
Vidal, J. P. O., et al. (2024). Enhancing extrusion die design efficiency through high-performance computing based optimization. Meccanica, 1-12.

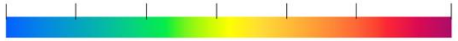
hpc4PE Case studies

Optimization in HPC systems

SOPREFA

		F_{obj} Improvement	Velocity Distribution					
ID	Material		Initial	Final	Cores	Trials	Time (Hours)	Cost (€)
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	.45
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	.77
P3	HDPE	2.3			512	409	23	115

- Reduction of 30-40% on the product time to market
- Reduction at least in 40% of the raw materials spent in experimental trials
- Estimated savings of 23 % in tools development



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EuroHPC Joint Undertaking